

# Worldsheet Conventions for Type IIB String Theory

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## Abstract

These notes fix the holomorphic worldsheet conventions used below for flat ten-dimensional type IIB string theory. The normalization of the free fields, the ghost systems, the BRST current, and the picture-changing operator follows Ref. [1]. In particular, the picture-changing operator is twice the normalization often used in the literature. The ten-dimensional gamma-matrix orientation and the normalization of the Ramond spin-field operator products follow Ref. [2]. Every displayed holomorphic formula has an anti-holomorphic counterpart obtained by placing a tilde on each field and replacing  $z, w$  by  $\bar{z}, \bar{w}$ .

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# 1 Flat-space worldsheet conventions

On each worldsheet coordinate chart, the Euclidean worldsheet coordinates are denoted by  $z$  and  $\bar{z}$ . Holomorphic fields depend on  $z$ , and anti-holomorphic fields depend on  $\bar{z}$ . Target-space vector indices  $M, N$  take values in  $\{0, 1, \dots, 9\}$ . Repeated target-space indices are summed over this set. We use the ten-dimensional Minkowski metric

$$\eta_{MN} := \text{diag}(-1, +1, \dots, +1). \quad (1.1)$$

Indices on target-space vectors are raised and lowered with  $\eta_{MN}$  and its inverse  $\eta^{MN}$ . We set

$$\alpha' = 1. \quad (1.2)$$

For two holomorphic operators  $A(z)$  and  $B(w)$ , the notation  $A(z)B(w) \sim C(z, w)$  means that, in the expansion valid for  $|z| > |w|$ ,  $C(z, w)$  contains precisely the terms that are singular as  $z \rightarrow w$ . Write  $\text{Sing}_{z \rightarrow w}[A(z)B(w)]$  for the sum of those singular terms. Normal ordering is defined by point splitting:

$$:AB:(w) := \lim_{z \rightarrow w} (A(z)B(w) - \text{Sing}_{z \rightarrow w}[A(z)B(w)]). \quad (1.3)$$

If  $T$  is a stress tensor, a holomorphic operator  $\mathcal{O}_h$  has conformal weight  $h$  when

$$T(z)\mathcal{O}_h(w) \sim \frac{h\mathcal{O}_h(w)}{(z-w)^2} + \frac{\partial\mathcal{O}_h(w)}{z-w}. \quad (1.4)$$

The central charge  $c$  of  $T$  is defined so that the coefficient of  $(z-w)^{-4}$  in  $T(z)T(w)$  is  $c/2$ . A field of Grassmann parity  $|A| \in \{0, 1\}$  is called even when  $|A| = 0$  and odd when  $|A| = 1$ ; away from operator-product singularities, two such fields obey  $AB = (-1)^{|A||B|}BA$ . Derivatives without a bar mean  $\partial := \partial/\partial z$ .

## 1.1 Conventions for the free boson

Let  $X^M(z, \bar{z})$  be the ten embedding coordinates. Their two-point operator product and its differentiated holomorphic form are

$$X^M(z, \bar{z})X^N(w, \bar{w}) \sim -\frac{\eta^{MN}}{2} \log|z-w|^2, \quad (1.5)$$

$$\partial X^M(z)\partial X^N(w) \sim -\frac{\eta^{MN}}{2(z-w)^2}. \quad (1.6)$$

Equation (1.6) is the free-boson normalization used in Ref. [1]. The holomorphic stress tensor of the ten bosons is defined by

$$T_X(z) := -\eta_{MN} : \partial X^M \partial X^N : (z). \quad (1.7)$$

With Eqs. (1.6) and (1.7), each  $X^M$  has holomorphic conformal weight zero, each  $\partial X^M$  has holomorphic conformal weight one, and the ten-boson central charge is

$$c_X = 10. \quad (1.8)$$

For a target-space momentum  $k_M \in \mathbb{R}^{1,9}$ , define

$$k \cdot X := k_M X^M, \quad (1.9)$$

$$k^2 := \eta^{MN} k_M k_N, \quad (1.10)$$

$$V_k(z, \bar{z}) := \exp(i k \cdot X(z, \bar{z})) : . \quad (1.11)$$

The holomorphic and anti-holomorphic conformal weights of  $V_k$  are

$$(h_k, \bar{h}_k) = \left( \frac{k^2}{4}, \frac{k^2}{4} \right). \quad (1.12)$$

Two useful consequences of Eq. (1.5) are

$$\partial X^M(z) V_k(w, \bar{w}) \sim -\frac{i k^M}{2(z-w)} V_k(w, \bar{w}), \quad (1.13)$$

$$V_k(z, \bar{z}) V_p(w, \bar{w}) \sim |z-w|^{k \cdot p} : \exp(i(k+p) \cdot X(w, \bar{w})) :, \quad (1.14)$$

where  $k \cdot p := \eta^{MN} k_M p_N$ .

## 1.2 Conventions for the free fermion

Let  $\psi^M(z)$  and  $\tilde{\psi}^M(\bar{z})$  be the holomorphic and anti-holomorphic components of the ten real worldsheet fermions. Their normalization is

$$\psi^M(z) \psi^N(w) \sim -\frac{\eta^{MN}}{2(z-w)}. \quad (1.15)$$

The holomorphic fermion stress tensor is

$$T_\psi(z) := \eta_{MN} : \psi^M \partial \psi^N : (z). \quad (1.16)$$

Thus  $\psi^M$  has holomorphic conformal weight  $1/2$ , and the ten-fermion central charge is

$$c_\psi = 5. \quad (1.17)$$

The free matter stress tensor and the matter supercurrent are defined by

$$T_m := T_X + T_\psi, \quad (1.18)$$

$$T_F := -\eta_{MN} : \psi^M \partial X^N : . \quad (1.19)$$

The minus sign in Eq. (1.19) is fixed jointly by Eqs. (1.6) and (1.15). It ensures that the picture-changing operator defined in Eq. (1.66) maps the  $-1$ -picture factor  $e^{-\phi}\psi^M$  to  $\partial X^M$  with positive coefficient. The superconformal operator products of  $T_m$  and  $T_F$  are derived in Sec. 1.5.

We next fix the gamma matrices and the Ramond spin fields. Spinor indices  $\alpha, \beta, \gamma$  take values in  $\{1, \dots, 16\}$ . We use symmetric chiral gamma-matrix blocks

$$(\Gamma^M)_{\alpha\beta} = (\Gamma^M)_{\beta\alpha}, \quad (\Gamma^M)^{\alpha\beta} = (\Gamma^M)^{\beta\alpha}. \quad (1.20)$$

Their normalization is defined by

$$(\Gamma^M)_{\alpha\beta}(\Gamma^N)^{\beta\gamma} + (\Gamma^N)_{\alpha\beta}(\Gamma^M)^{\beta\gamma} = 2\eta^{MN}\delta_\alpha^\gamma. \quad (1.21)$$

Equivalently, the  $32 \times 32$  matrices

$$\mathbf{\Gamma}^M := \begin{pmatrix} 0 & (\Gamma^M)^{\alpha\beta} \\ (\Gamma^M)_{\alpha\beta} & 0 \end{pmatrix} \quad (1.22)$$

satisfy

$$\{\mathbf{\Gamma}^M, \mathbf{\Gamma}^N\} = 2\eta^{MN}\mathbf{1}_{32}. \quad (1.23)$$

For  $p \geq 1$ , the antisymmetrized product is

$$\mathbf{\Gamma}^{M_1 \dots M_p} := \mathbf{\Gamma}^{[M_1} \dots \mathbf{\Gamma}^{M_p]}, \quad (1.24)$$

where the brackets include the factor  $1/p!$ . A symbol such as  $(\Gamma^{M_1 \dots M_p})_\alpha^\beta$  denotes the indicated chiral block of Eq. (1.24). We choose the target-space orientation and positive-chirality block by

$$\epsilon^{01 \dots 9} = +1, \quad (\Gamma^{01 \dots 9})_\alpha^\beta = \delta_\alpha^\beta. \quad (1.25)$$

Equations (1.20) and (1.25) are the ten-dimensional conventions of Ref. [2].

Let  $S_\alpha(z)$  and  $\tilde{S}_\alpha(\bar{z})$  denote the left- and right-moving positive-chirality Ramond spin fields.

Type IIB means that these two fields have the same ten-dimensional chirality. Each has conformal weight  $5/8$  in its chiral sector. The symbols  $S^\alpha$  and  $\tilde{S}^\alpha$  denote the spin fields that occur in the  $-3/2$  picture; the placement of the spinor index is part of this convention and is fixed by the operator products below, rather than by an unstated index-raising rule.

The Ramond operator products also contain the bosonized superghost  $\phi$ . For use in the next three equations,  $\phi$  is defined by

$$\partial\phi(z)\partial\phi(w) \sim -\frac{1}{(z-w)^2}, \quad e^{q_1\phi}(z)e^{q_2\phi}(w) \sim (z-w)^{-q_1q_2}e^{(q_1+q_2)\phi}(w). \quad (1.26)$$

The complete  $\beta\gamma$  bosonization is given in Sec. 1.4. We normalize the spin fields by the singular terms

$$e^{-\phi}\psi^M(z) e^{-\phi/2}S_\alpha(w) \sim \frac{i}{2(z-w)}(\Gamma^M)_{\alpha\beta}e^{-3\phi/2}S^\beta(w), \quad (1.27)$$

$$e^{-3\phi/2}S^\alpha(z) e^{-\phi/2}S_\beta(w) \sim \frac{\delta^\alpha_\beta}{(z-w)^2}e^{-2\phi}(w), \quad (1.28)$$

$$e^{-\phi/2}S_\alpha(z) e^{-\phi/2}S_\beta(w) \sim \frac{i}{z-w}(\Gamma^M)_{\alpha\beta}e^{-\phi}\psi_M(w). \quad (1.29)$$

The branch choices, phases, and factors in Eqs. (1.27)–(1.29) are those of Ref. [2]. The anti-holomorphic spin fields obey the same equations with every field and coordinate replaced by its anti-holomorphic counterpart. This identical left-right form is another place where the type IIB chirality choice enters.

### 1.3 Conventions for the $bc$ ghost system

The reparametrization ghosts  $b(z)$  and  $c(z)$  are Grassmann odd. Their holomorphic conformal weights, ghost numbers, and singular operator product are

|       |     |              |                                |        |
|-------|-----|--------------|--------------------------------|--------|
| field | $h$ | ghost number | $b(z)c(w) \sim \frac{1}{z-w}.$ | (1.30) |
| $b$   | 2   | −1           |                                |        |
| $c$   | −1  | +1           |                                |        |

The  $bc$  stress tensor is defined by

$$T_{bc} := -2 :b\partial c: - :(\partial b)c:, \quad (1.31)$$

and its central charge is

$$c_{bc} = -26. \quad (1.32)$$

On the complex plane, a nonzero  $bc$  correlator contains three more  $c$  insertions than  $b$  insertions. We define the constant  $K$  by the normalization used in Ref. [1]:

$$\langle c(z_1)c(z_2)c(z_3) \rangle_{bc} := -K(z_1 - z_2)(z_2 - z_3)(z_1 - z_3). \quad (1.33)$$

Here the subscript  $bc$  means that the expectation value is taken only in the  $bc$  sector. Equation (1.33) defines  $K$ ; its relation to a D-brane or D-instanton tension additionally depends on the normalization of the string amplitude and is not assumed in these notes.

## 1.4 Conventions for the $\beta\gamma$ system

The superconformal ghosts  $\beta(z)$  and  $\gamma(z)$  are Grassmann even and have conformal weights  $3/2$  and  $-1/2$ , respectively. We bosonize them in terms of Grassmann-odd fields  $\xi, \eta$  and a Grassmann-even chiral scalar  $\phi$  by

$$\beta := \partial\xi e^{-\phi}, \quad (1.34)$$

$$\gamma := \eta e^{\phi}. \quad (1.35)$$

The order of the factors in Eqs. (1.34) and (1.35) is part of the convention. The elementary operator products are

$$\xi(z)\eta(w) \sim \frac{1}{z-w}, \quad (1.36)$$

$$\partial\phi(z)\partial\phi(w) \sim -\frac{1}{(z-w)^2}, \quad (1.37)$$

$$e^{q_1\phi}(z)e^{q_2\phi}(w) \sim (z-w)^{-q_1q_2}e^{(q_1+q_2)\phi}(w). \quad (1.38)$$

It follows from Eqs. (1.34)–(1.38) that

$$\beta(z)\gamma(w) \sim -\frac{1}{z-w}, \quad \gamma(z)\beta(w) \sim \frac{1}{z-w}. \quad (1.39)$$

The bosonized stress tensors are

$$T_{\xi\eta} := -:\eta\partial\xi:, \quad (1.40)$$

$$T_\phi := -\frac{1}{2}:(\partial\phi)^2: -\partial^2\phi, \quad (1.41)$$

$$T_{\beta\gamma} := T_{\xi\eta} + T_\phi. \quad (1.42)$$

In the unbosonized variables, the same stress tensor is

$$T_{\beta\gamma} = -\frac{3}{2} : \beta \partial \gamma : - \frac{1}{2} : (\partial \beta) \gamma : . \quad (1.43)$$

Its central charge is

$$c_{\beta\gamma} = 11. \quad (1.44)$$

The ghost number and picture number used below are additive charges defined on the elementary bosonized fields by

| field          | $\beta$ | $\gamma$ | $\xi$ | $\eta$ | $e^{q\phi}$ |   |
|----------------|---------|----------|-------|--------|-------------|---|
| ghost number   | -1      | +1       | -1    | +1     | 0           | . |
| picture number | 0       | 0        | +1    | -1     | $q$         |   |

(1.45)

The conformal weight of the exponential is

$$h[e^{q\phi}] = -\frac{1}{2}q(q+2). \quad (1.46)$$

The mode expansions of the weight-zero field  $\xi$  and the weight-one field  $\eta$  define

$$\xi_0 := \oint \frac{dz}{2\pi i z} \xi(z), \quad \eta_0 := \oint \frac{dz}{2\pi i} \eta(z). \quad (1.47)$$

The large Hilbert space is the state space on which  $\xi_0$  acts without the restriction in Eq. (1.48).

The small Hilbert space is the subspace of states  $|\Psi\rangle$  satisfying

$$\eta_0 |\Psi\rangle = 0. \quad (1.48)$$

Equations (1.34)–(1.48) follow the ordering, charges, and Hilbert-space convention of Ref. [1].

## 1.5 Conventions for the $TT$ and $T_F T_F$ operator products

Let  $T(z)$  be the stress tensor of a holomorphic matter superconformal field theory, let  $T_F(z)$  be its supercurrent in the normalization defined below, and let  $c_m$  be its central charge. We

define the normalization of  $T$  and  $T_F$  by the singular operator products

$$T(z)T(w) \sim \frac{c_m/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}, \quad (1.49)$$

$$T(z)T_F(w) \sim \frac{3T_F(w)}{2(z-w)^2} + \frac{\partial T_F(w)}{z-w}, \quad (1.50)$$

$$T_F(z)T_F(w) \sim \frac{c_m/6}{(z-w)^3} + \frac{T(w)}{2(z-w)}. \quad (1.51)$$

Thus  $T_F$  is one half of a supercurrent  $G$  normalized by  $G(z)G(w) \sim (2c_m/3)(z-w)^{-3} + 2T(w)(z-w)^{-1}$ . This explicit algebraic relation is the definition of the relative normalization; no separate meaning is assigned to the phrase “normalized supercurrent.” Equations (1.49)–(1.51) match the convention used in Ref. [1].

For the ten free bosons and ten free fermions of Secs. 1.1 and 1.2,

$$c_m = c_X + c_\psi = 15, \quad T = T_m. \quad (1.52)$$

The operator products therefore become

$$T_m(z)T_m(w) \sim \frac{15/2}{(z-w)^4} + \frac{2T_m(w)}{(z-w)^2} + \frac{\partial T_m(w)}{z-w}, \quad (1.53)$$

$$T_m(z)T_F(w) \sim \frac{3T_F(w)}{2(z-w)^2} + \frac{\partial T_F(w)}{z-w}, \quad (1.54)$$

$$T_F(z)T_F(w) \sim \frac{5/2}{(z-w)^3} + \frac{T_m(w)}{2(z-w)}. \quad (1.55)$$

For completeness, the coefficients in Eq. (1.55) can be reproduced directly. Substituting Eq. (1.19), the double contraction of the two fermions and the two differentiated bosons gives

$$\left(-\frac{\eta^{MN}}{2(z-w)}\right) \left(-\frac{\eta_{MN}}{2(z-w)^2}\right) = \frac{10}{4(z-w)^3} = \frac{5}{2(z-w)^3}. \quad (1.56)$$

The single fermion contraction gives

$$-\frac{1}{2(z-w)} : \partial X^M \partial X_M : (w) = \frac{T_X(w)}{2(z-w)}. \quad (1.57)$$

For the single boson contraction, expand  $\psi^M(z) = \psi^M(w) + (z-w)\partial\psi^M(w) + \mathcal{O}((z-w)^2)$ . The zeroth-order normal product  $:\psi^M\psi_M:$  vanishes, while the first-order term gives

$$-\frac{1}{2(z-w)} : (\partial\psi^M)\psi_M : (w) = \frac{T_\psi(w)}{2(z-w)}. \quad (1.58)$$



Adding Eqs. (1.56)–(1.58) gives Eq. (1.55) with  $T_m = T_X + T_\psi$ .

The total holomorphic stress tensor of the gauge-fixed critical theory is

$$T_{\text{tot}} := T_m + T_{bc} + T_{\beta\gamma}. \quad (1.59)$$

Its central charge vanishes because

$$c_{\text{tot}} = c_m + c_{bc} + c_{\beta\gamma} = 15 - 26 + 11 = 0. \quad (1.60)$$

## 1.6 Conventions for the BRST and PCO operators

The holomorphic BRST current is defined by

$$j_B := c(T_m + T_{\xi\eta} + T_\phi) + bc\partial c + \gamma T_F - \frac{1}{4}\gamma^2 b. \quad (1.61)$$

All products on the right-hand side of Eq. (1.61) are normal ordered. In bosonized variables, the composite field  $\gamma^2$  is

$$\gamma^2 := :\gamma\gamma: = \eta\partial\eta e^{2\phi}. \quad (1.62)$$

The BRST charge is the contour integral

$$Q_B := \oint \frac{dz}{2\pi i} j_B(z), \quad (1.63)$$

where the contour is counterclockwise around the operator on which  $Q_B$  acts. The balance  $c_m = 15$  in Eq. (1.52) implies

$$Q_B^2 = 0. \quad (1.64)$$

The anti-holomorphic charge  $\tilde{Q}_B$  is defined by the tilded copy of Eqs. (1.61)–(1.63); the closed-string BRST charge is

$$Q_{\text{closed}} := Q_B + \tilde{Q}_B. \quad (1.65)$$

For two Grassmann-odd operators  $A$  and  $B$ , write  $\{A, B\} := AB + BA$ . The holomorphic picture-changing operator (PCO) is defined in the large Hilbert space by

$$\mathcal{X}(z) := 2\{Q_B, \xi(z)\} \quad (1.66)$$

$$= 2c\partial\xi + 2e^\phi T_F - \frac{1}{2}[(\partial\eta)e^{2\phi}b + \partial(\eta e^{2\phi}b)](z). \quad (1.67)$$

The factor of two in Eq. (1.66) is part of the convention adopted from Ref. [1]. Define  $X_{\text{common}} := \{Q_B, \xi\}$  to be the PCO without this factor. The two conventions obey

$$\mathcal{X} = 2X_{\text{common}}. \quad (1.68)$$

The PCO  $\mathcal{X}$  has conformal weight zero, ghost number zero, picture number +1, and is BRST closed:

$$[Q_B, \mathcal{X}(z)] = 0. \quad (1.69)$$

The anti-holomorphic PCO is

$$\tilde{\mathcal{X}}(\bar{z}) := 2\{\tilde{Q}_B, \tilde{\xi}(\bar{z})\}. \quad (1.70)$$

For an oriented worldsheet of genus  $g \in \mathbb{Z}_{\geq 0}$  with  $b \in \mathbb{Z}_{\geq 0}$  boundary components, define  $P_{\text{tot}}$  to be the sum of the picture numbers of all holomorphic and anti-holomorphic insertions, including PCOs. A nonzero correlator requires

$$P_{\text{tot}} = 2(2g + b - 2). \quad (1.71)$$

For a closed worldsheet,  $b = 0$ , this condition can be imposed separately as  $P_L = P_R = 2g - 2$ . Equation (1.71) explains why PCO insertions are required when the picture numbers of the vertex operators do not already have the required sum.

## References

- [1] Jyotirmoy Barman, Rishabh Kaushik, Raghu Mahajan, Chitraang Murdia, and Ashoke Sen, “Instanton-Induced Closed-String Amplitudes in Minimal Superstring Theory at Subleading Order,” (2026), [arXiv:2606.06596 \[hep-th\]](#)
- [2] Sergei Alexandrov, Ashoke Sen, and Bogdan Stefański, “Euclidean D-branes in Type IIB string theory on Calabi-Yau threefolds,” [JHEP](#) **12**, 044 (2021), [arXiv:2110.06949 \[hep-th\]](#)